

IB-003 Results — Intent-Binding Pipeline Retest With Authority Normalization

Experiment ID: IB-003

Parent experiments: IB-001 (FAIL — K2), IB-002 (FAIL — K3)

Series: Intent Binding (IB)

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Framework: Coherent Inheritance Framework (CIF) / ExecutionProof

Preregistration hash: d6f788f46c05da853a2ebaefa59ef1c69863e3d3d810d44c739c64dae5fb7d8b

ProofRecord hash: 9862b782777e0866dcfaefce49096369ad8d04aa19709b99686cebb7e0ad107d

Verdict: PASS (strong positive)

12/12 correct. 0 false holds. 8/8 drift caught. 0 privilege expansions. All 4 kill conditions clear.

The Complete Sequence

Experiment	Verdict	What it found
IB-001	FAIL (K2)	Intent-binding pipeline exposed an authority-layer synonym gap. IntentMatch correctly classified all 12 cases in that test set; the pipeline failed because <code>dis-patch_email</code> and <code>view_file</code> were rejected by authority before IntentMatch ran.
IB-002	FAIL (K3)	Authority normalization solved the alias problem without fuzzy expansion. Within the 18 preregistered IB-002 cases, the normalization function produced zero observed privilege expansions, resolved all six legitimate aliases, and rejected all six near-neighbor attacks. The experiment also exposed a test-design conflation between normalization safety and authority-grant breadth.
IB-003	PASS (strong positive)	The remediated pipeline — authority normalization integrated, everything else unchanged — achieves 12/12 on the original IB-001 cases. The two false blocks (C02, C03) are resolved. All 8 drift detections preserved.

This sequence is stronger than a single clean PASS because the system was allowed to fail twice, each failure was preserved, the root cause was isolated, and the remediation was tested independently before reintegration.

Results: Side-by-Side Comparison with IB-001

Controls (expected ALLOW)

Case	Description	IB-001 Cond B	IB-003 Cond B	Changed
C01	Exact intended action	ALLOW ✓	ALLOW ✓	No
C02	Harmless wording (<code>dispatch_email</code>)	DENY ✗	ALLOW ✓	Yes
C03	Nonmaterial metadata (<code>view_file</code>)	DENY ✗	ALLOW ✓	Yes
C04	Approved payment, correct party/amount	ALLOW ✓	ALLOW ✓	No

Controls correct: IB-001 = 2/4 → IB-003 = 4/4

The two false blocks were caused by the authority layer rejecting `dispatch_email` and `view_file` before `IntentMatch` could evaluate them. With `normalize_authority_action_type()` applied, both synonyms resolve to their canonical forms (`send_email` , `read_file`) and pass authority. `IntentMatch` then confirms zero material divergences on both.

Drift Cases (expected HOLD)

Case	Description	IB-001 Cond B	IB-003 Cond B	Changed	Drift Axis
C05	Recipient changed	HOLD ✓	HOLD ✓	No	target
C06	Amount increased	HOLD ✓	HOLD ✓	No	amount
C07	Recipient silently added	HOLD ✓	HOLD ✓	No	prohibited_side_effect
C08	Read → write	HOLD ✓	HOLD ✓	No	action_type
C09	External transmission	HOLD ✓	HOLD ✓	No	prohibited_side_effect
C10	Single → batch	HOLD ✓	HOLD ✓	No	scope, prohibited_side_effect
C11	Tool substitution	HOLD ✓	HOLD ✓	No	action_type, prohibited_side_effect
C12	Different objective	HOLD ✓	HOLD ✓	No	purpose, prohibited_side_effect

Drift caught: IB-001 = 8/8 → IB-003 = 8/8

No regressions. Every drift case that IB-001's IntentMatch caught remains caught.

Aggregate Comparison

Metric	IB-001	IB-003	Change
Condition A accuracy	2/12	4/12	+2 (authority now passes C02, C03)
Condition B accuracy	10/12	12/12	+2 (C02, C03 resolved)
Controls correct (B)	2/4	4/4	+2
Drift caught (B)	8/8	8/8	Unchanged
False holds (B)	2	0	-2
Kill triggered	K2	None	Resolved
Verdict	FAIL	PASS (strong positive)	

Kill Condition Evaluation

Kill Condition	Status	Detail
K1 — Drift regression	✓ Clear	0 previously caught drift cases escaped
K2 — Control still blocked	✓ Clear	0 controls remain incorrectly blocked
K3 — Scope contamination	✓ Clear	Only <code>check_authority()</code> was modified; all other code identical to IB-001
K4 — Privilege expansion	✓ Clear	0 privilege expansions detected

Privilege Expansion Audit

Case	Original	Normalized To	Authorized	Legitimate Synonym
C02	dispatch_email	send_email	Yes	Yes (declared)
C03	view_file	read_file	Yes	Yes (declared)

All normalizations are declared legitimate synonyms from the preregistered `WORDING_EQUIVALENCES` map. No privilege expansions detected.

What Condition A Shows

Condition A (authority + evidence, no IntentMatch) improved from 2/12 to 4/12 because C02 and C03 now pass authority. But Condition A still allows all 8 drift cases through — it has no mechanism to detect semantic drift when authority and evidence are valid.

This confirms the IB-001 finding: authority and evidence are necessary but insufficient. IntentMatch provides a distinct, measurable security layer.

Scientific Value of the Full Sequence

1. **The system was allowed to fail.** IB-001 FAIL is preserved. IB-002 FAIL is preserved. Neither was retroactively softened.
2. **Each failure isolated a distinct structural issue.** IB-001 found the authority-synonym gap. IB-002 found the test-design conflation between normalization safety and authority-grant breadth.
3. **The remediation was tested independently before reintegration.** IB-002 confirmed the normalization function resolves aliases (6/6), rejects near-neighbors (6/6), and produces zero privilege expansions — within those 18 preregistered cases.
4. **The retest used the original ground-truth labels.** No labels were changed. No IntentMatch rules were modified. No evidence or authority grant was altered. The only change was the one preregistered remediation.
5. **The result is bounded, not universal.** 12/12 on this preregistered test set with 4 controls and 8 handcrafted drift cases. The next adversarial rounds should test ambiguous scope, nested actions, conflicting constraints, chained tools, temporal drift, and partial equivalence.

Remaining Open Questions (Not Addressed by IB-003)

1. **Authority-grant breadth (AUTH-001 direction).** IB-002 exposed that the flat authority grant includes high-privilege actions alongside low-privilege ones. Whether separating `send_email`,

`read_file`, `delete_record`, `execute_tool`, `batch_execute`, and `transmit_external` into narrower capability grants reduces unintended authorization is a separate research question.

2. **Adversarial scale.** 12 cases is sufficient for a first demonstration. It is not sufficient to claim the IntentMatch primitive is generally robust. Future experiments should attempt to break it with harder cases.
3. **Natural-language intent extraction.** All experiments so far use a pre-normalized IntentContract. Whether intent can be reliably extracted from natural-language authorization in production environments is a separate and harder problem.

Preserved Summary Sentences

IB-001:

“IB-001 failed at the pipeline level while isolating a distinct upstream authority-normalization defect; the IntentMatch primitive correctly classified all 12 preregistered cases in this test set.”

IB-002:

“IB-002 failed at the experiment level (K3 triggered) while demonstrating that, within the 18 preregistered IB-002 cases, the normalization function produced zero observed privilege expansions, resolved all six legitimate aliases, and rejected all six near-neighbor attacks; the failure isolated a test-design conflation between normalization safety and authority-grant breadth.”

IB-003:

“IB-003 passed with a strong positive result (12/12) after applying the preregistered authority-normalization remediation to the original IB-001 pipeline; the two false blocks (C02, C03) were resolved, all eight semantic-drift detections were preserved, and zero privilege expansions were detected — within this preregistered test set of 12 cases.”

Files

File	SHA-256
PREREGISTRATION.md	d6f788f46c05da853a2ebaefa59e- f1c69863e3d3d810d44c739c64dae5fb7d8b
run_ib003.py	c1f3ca2d27bd0c5538b00babd620bc0d- f86012eda41545ff5c7ba51064f0b306
results/ib003_results.json	c0eb- c1058532ba4e05013243d1196e5d1cec4d415c367c0 8cbe8512d5a059d26
results/proofrecord_ib003.json	699edb4253dd1a9bac91e3219d53e50abf80e2c9bd- d1df40f2a9ca223320e562
results/ib003_ledger.jsonl	abcc512a40793b4945676c688675a87f3fc0f83d6- bea4ea29cd440503d79bd3c